Emergent Symbol Systems and Cultural Memory in a Self-Organizing Artificial Civilization

Sanchay Kumar
ysanchay@gmail.com
https://github.com/ysanchay/civilization-simulator

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Abstract

We present an experimental framework for studying the emergence of symbolic systems, cultural memory, and civilizational dynamics in decentralized multi-agent systems. Our simulator implements tribes of agents that inhabit a dynamic grid world, form symbols from perceived patterns, and develop predictive, causal, and goal-oriented representations. We demonstrate that language composition emerges spontaneously when symbol vocabularies exceed a transition threshold, and meta-symbol abstraction forms through co-occurrence tracking across tribes. The system implements cultural forgetting and compression mechanisms that drive abstraction under memory pressure. Critically, we measure knowledge independently of behavior through artifact structures—knowledge-only objects that agents can observe but that provide no direct survival reward. Our experiments reveal four key empirical findings: (1) war frequency follows a heavy-tailed distribution with kurtosis 11.59, indicating clustered conflict dynamics; (2) collapse events cluster into four distinct archetypes with different causal signatures; (3) symbolic complexity exhibits open-ended growth with a 4.62x increase over 1500 simulation steps; and (4) knowledge retention across simulation runs averages 41%, improving with repeated exposure. We introduce three theoretical contributions: the Cultural Compression Hypothesis (memory pressure drives abstraction), the Predictive Survival Hypothesis (temporal prediction correlates with survival), and the Symbol Competition Hypothesis (meta-symbols replace base representations). These findings contribute to our understanding of emergent complexity in multi-agent systems and provide a controlled testbed for AI safety research on value emergence and civilizational stability.

Keywords: multi-agent systems, emergent behavior, symbolic grounding, cultural evolution, civilizational collapse, AI safety, language emergence, abstraction

1 Introduction

1.1 Motivation

How do civilizations emerge? This question has fascinated philosophers, historians, and scientists for centuries. Recent advances in multi-agent systems and artificial intelligence provide new tools for empirical investigation. By constructing artificial societies with minimal assumptions, we can observe emergent phenomena and test hypotheses about civilizational dynamics.

This work presents a civilization simulator designed to study:

1. **Symbol Grounding**: How do abstract symbols emerge from grounded perception?
2. **Cultural Memory**: How is knowledge preserved and transmitted across generations?
3. **Civilizational Collapse**: What causes societies to fail, and can we predict it?
4. **Open-Ended Evolution**: Does complexity grow indefinitely or saturate?

1.2 Why Current RL Systems Lack Cultural Continuity

Standard reinforcement learning systems suffer from fundamental limitations when it comes to cultural evolution:

- **Fixed Reward Functions**: RL agents optimize predetermined objectives, precluding the emergence of novel goals or values
- **No Memory Persistence**: Each episode typically resets state, eliminating the possibility of cumulative cultural knowledge
- **Isolation**: Agents learn independently without mechanisms for knowledge transfer
- **No Abstraction**: Representations remain at the level of sensory input, without symbol formation or composition

These limitations make standard RL unsuitable for studying how cultures form, evolve, and transmit knowledge across generations.

1.3 Why Symbol Emergence Matters

Symbol systems are the foundation of intelligent behavior:

- **Compression**: Symbols allow efficient representation of complex patterns
- **Communication**: Shared symbols enable coordination and knowledge transfer
- **Reasoning**: Abstract symbols support planning and prediction
- **Culture**: Symbol systems persist across generations, forming cultural memory

Understanding how symbols emerge from raw perception—without being programmed—is essential for building AI systems that can develop their own representations and goals.

1.4 Why Language and Abstraction Are Key

Language and abstraction are intimately connected:

- **Composition**: Language allows combining symbols to express novel concepts
- **Meta-levels**: Abstraction enables reasoning about symbols themselves
- **Transfer**: Abstract representations transfer across domains
- **Compression**: Higher-level symbols compress multiple lower-level patterns

Our simulator demonstrates that language emerges spontaneously when symbol vocabularies grow large enough, and that meta-symbols form through cultural competition.

1.5 Why Multi-Tribe Transfer Is Important

Cultural evolution depends on knowledge transfer between groups:

- **Diffusion**: Ideas spread between tribes through contact
- **Competition**: Better representations outcompete worse ones
- **Specialization**: Different tribes develop different expertise
- **Resilience**: Distributed knowledge survives individual tribe collapses

Our simulator implements inter-tribe knowledge transfer through symbol adoption, creating a genuine cultural evolution dynamic.

1.6 Core Hypothesis

Our central hypothesis is that **intelligence can emerge from survival pressure combined with memory, compression, and inter-tribe transfer**. This hypothesis suggests that:

1. Agents under survival pressure will form symbols for patterns affecting survival
2. Memory constraints will drive abstraction (compression)
3. Inter-tribe transfer will accelerate cultural evolution
4. Meta-symbols will emerge to represent common symbol patterns

We test this hypothesis through controlled experiments that vary each factor independently.

1.7 Contributions

We make the following contributions:

- A complete simulation framework implementing territorial dynamics, cultural evolution, cognitive stress, and collapse mechanics
- A three-level symbol system demonstrating emergence, composition, and meta-abstraction
- Cultural memory and forgetting mechanisms that drive abstraction under pressure
- Inter-tribe knowledge transfer enabling cultural evolution
- Knowledge metrics measured independently of behavior through artifact structures
- Empirical validation of four key phenomena: heavy-tailed war distribution, collapse archetypes, open-ended complexity growth, and knowledge persistence
- Three theoretical frameworks: Cultural Compression, Predictive Survival, and Symbol Competition hypotheses

1.8 Scope

We explicitly position this work as an *experimental framework* rather than a model of actual human civilization. Our agents are simplified abstractions, and our findings should be interpreted as hypotheses about possible dynamics rather than claims about historical processes.

Important Constraint: Symbol grounding in this framework is *limited to survival-relevant perceptions*. Agents develop symbols only for aspects of their environment that affect survival—resource locations, threats, territory boundaries, and tribal identity. This constraint increases the ecological validity of the model, as early symbolic systems in biological evolution similarly emerged from survival pressures rather than abstract reasoning.

2 Related Work

2.1 Agent-Based Social Simulation

Agent-based modeling has been applied to social systems since Schelling's segregation model [1]. Recent work includes:

- **SugarScape** [2]: Artificial societies with resource dynamics
- **Cultural Evolution** [3]: Dissemination of culture through social influence

- **Artificial Anasazi** [4]: Archaeological simulation of societal collapse

Our work extends these traditions with explicit symbol systems and cultural memory mechanisms.

2.2 Emergent Communication

Research on emergent communication in multi-agent systems has demonstrated:

- **Referential Games** [5]: Agents develop communication protocols
- **Grounded Language Learning** [6]: Language emerges from embodied interaction
- **Symbol Systems** [7]: Agents develop shared vocabularies

We build on this by studying symbol *abstraction* and *cultural transmission* across generations.

2.3 Symbol Grounding

The symbol grounding problem [13] asks how symbols acquire meaning. Our framework addresses this through:

- Survival-grounded perception: Symbols form from patterns affecting survival
- Reward grounding: Symbol values derive from survival outcomes
- Causal role accumulation: Symbols gain meaning through use in prediction and planning

2.4 Civilizational Collapse

Historical collapse studies [8, 9] identify multiple collapse causes:

- Resource depletion
- Complexity overload
- External invasion
- Internal conflict

Our simulator implements these as mechanisms, allowing controlled experiments.

2.5 Comparison to Alternative Approaches

Our framework differs from related paradigms in important ways:

Reinforcement Learning: Standard RL assumes fixed reward functions and lacks mechanisms for emergent goals or cultural transmission. Our agents develop their own value systems through symbol formation.

Evolutionary Algorithms: While EA supports open-ended complexity growth, it typically lacks the cultural transmission and memory mechanisms we implement.

Cultural Evolution Models: These models focus on trait transmission but rarely implement full civilizational dynamics including collapse, territorial control, and cognitive stress.

Table 1: Comparison to Related Paradigms

Feature	Ours	RL	Evo. Alg.	Cult. Evo.
Emergent goals	Yes	No	Partial	Partial
Symbol grounding	Yes	No	No	Limited
Cultural transmission	Yes	No	No	Yes
Collapse dynamics	Yes	No	No	No
Open-ended evolution	Yes	No	Yes	Partial
Memory persistence	Yes	No	Limited	Limited
Abstraction levels	Yes	No	No	No

2.6 AI Safety Implications

Multi-agent systems with emergent values relate to AI safety concerns:

- **Value Emergence** [10]: How do goals form in intelligent systems?
- **Multi-Agent Alignment** [11]: Coordination among AI systems
- **Memetics** [12]: Cultural units spread and evolve like genes

Our framework provides a testbed for studying value formation, goal drift, and misalignment dynamics in a controlled setting.

3 Environment Design

3.1 Grid World

The simulation environment is a two-dimensional grid world with the following characteristics:

Grid Structure:

- **Dimensions**: 20×20 to 30×30 cells (configurable)
- **Biomes**: Different terrain types with varying resource density
- **Food Regeneration**: Resources respawn at configurable rates
- **Edge Danger**: Boundary cells impose survival penalties
- **Crowd Pressure**: High population density reduces efficiency

Table 2: World Parameters

Parameter	Description	Default
GRID_SIZE	World dimensions	25×25
FOOD_SPAWN_RATE	Probability per cell per step	0.02
BIOME_COUNT	Number of terrain types	4
EDGE_DANGER	Damage at boundaries	0.1
CROWD_THRESHOLD	Density penalty trigger	5

3.2 Stochastic World Time

The simulator implements non-linear time progression to create environmental instability:

- **Epoch Shifts**: Periodic changes in world rules (resource distribution, danger zones)
- **Environmental Instability**: Random perturbations to terrain and resources
- **Temporal Chaos**: Irregular cycles that challenge prediction

This stochasticity tests agent adaptability and drives the formation of predictive symbols.

3.3 Artifacts: Knowledge-Only Structures

A critical innovation in our framework is the *artifact* system. Artifacts are objects that agents can observe but that provide **no direct survival reward**.

Artifact Types:

- **Counter**: Records observations over time
- **Cycle**: Represents periodic patterns
- **Conditional**: Encodes if-then relationships

Why Artifacts Matter: Artifacts allow us to measure knowledge *independently of behavior*. An agent that forms correct symbols for artifact patterns demonstrates genuine understanding, not just reward optimization. This is academically significant because:

1. It separates representation learning from policy learning
2. It provides an objective measure of abstraction capability
3. It tests whether agents form symbols for non-instrumental patterns
4. It enables measurement of knowledge transfer across domains

4 Agent Architecture

4.1 Physiology

Each agent (tribe) maintains physiological state:

$$P = \langle E, M, D, R \rangle \tag{1}$$

Where:

- E = Energy level $\in [0, 1]$
- M = Metabolism rate (energy consumption per step)
- D = Dominance pressure from neighboring tribes
- R = Reproduction threshold (energy level for splitting)

Energy Dynamics:

$$\frac{dE}{dt} = -M + \sum_{c \in T} \text{food}(c) - \text{stress}(c) \tag{2}$$

Where T is the tribe's territory and stress(c) is the environmental stress at cell c.

4.2 Cognition

Each agent maintains a belief state:

$$B = \langle S, M, T, \eta, \sigma \rangle \tag{3}$$

Where:

- S = Symbol dictionary $\{pattern \rightarrow value\}$
- M = Meta-symbols (composed symbols)
- T = Controlled territory (set of cells)
- η = Current efficiency $\in [0, 1]$
- σ = Stress level $\in [0, 1]$

Pattern Observation: Agents observe patterns from four sources:

1. **Food**: Resource locations and densities
2. **Danger**: Threat zones and damage rates
3. **Artifact**: Knowledge-only structures
4. **Time**: Temporal patterns and cycles

Planning Horizon: Agents simulate future states up to h steps ahead, where h is limited by cognitive stress:

$$h_{effective} = h_{base} \times (1 - \sigma) \tag{4}$$

Surprise Penalty: Unexpected observations increase stress:

$$\Delta\sigma = \gamma \cdot \mathcal{S}(o|B) \cdot (1 - \sigma) \tag{5}$$

4.3 Multi-Step Planning

Agents plan actions through recursive simulation:

```
def simulate_move(state, depth):
    if depth == 0:
        return evaluate(state)

    best_value = -infinity
    for action in possible_actions:
        next_state = transition(state, action)
        value = simulate_move(next_state, depth - 1)
        if value > best_value:
            best_value = value
            best_action = action

    return best_value, best_action
```

The planning depth is limited by stress and energy, creating a resource-rational decision process.

4.4 Cultural Communication

Tribes communicate through symbol sharing:

- **Symbol Exchange**: Neighboring tribes can observe each other's symbol dictionaries
- **Adoption Probability**: Symbols are adopted based on their demonstrated value
- **Competition**: Conflicting symbols compete for dominance

5 Symbol System

The symbol system is our core theoretical contribution. We implement a three-level hierarchy of symbol emergence.

5.1 Level 1: Raw Symbols

Symbols emerge from patterns in agent perception:

Pattern Extraction:

$$\text{Perception} \xrightarrow{\text{extract}} \text{Pattern} \xrightarrow{\text{create}} \text{Symbol} \xrightarrow{\text{assign}} \text{Value} \tag{6}$$

Reward Grounding: Symbol values are grounded in survival outcomes:

$$V(s) = \mathbb{E}[R|s \text{ observed}] \tag{7}$$

Where R is the reward (energy change) following observation of symbol s.

Causal Role Accumulation: Symbols gain additional meaning through use:

$$\text{CausalRole}(s) = \{(s', a, r) : s \rightarrow_a s' \rightarrow r\} \tag{8}$$

A symbol becomes predictive if it reliably precedes outcomes, causal if it enables actions that cause outcomes, and goal-oriented if it represents desired states.

Table 3: Symbol Roles

Role	Definition	Example
Predictive	$\mathbb{E}[R\|s] \neq \mathbb{E}[R]$	DANGER $\rightarrow$ damage
Causal	$P(s'\|s, a) > P(s'\|a)$	FARM $\rightarrow$ food (if expand)
Goal	$V(s) > V(s')$ for reachable s'	RICH $\rightarrow$ target state

5.2 Level 2: Composed Symbols (Language Emergence)

When the raw symbol vocabulary exceeds a threshold, composition becomes beneficial:

Transition Threshold: Let $|S|$ be the symbol count. When $|S| > \theta_{trans}$, agents begin forming composed symbols:

$$s_{composed} = s_i \circ s_j \quad \text{if } P(s_i, s_j) > \phi_{cooccur} \tag{9}$$

Where $P(s_i, s_j)$ is the co-occurrence probability and $\phi_{cooccur}$ is the composition threshold.

Language Emergence Event: The formation of the first composed symbol marks the transition from pre-linguistic to linguistic representation. This is a phase transition in the system:

$$\text{LanguageEmerges} = \mathbf{1}[|S_{composed}| > 0] \tag{10}$$

Table 4: Symbol Hierarchy

Level	Type	Grounding	Example
0	Primitive	Direct observation	FERTILE
1	Composed	Multiple primitives	FARM = FERTILE + WATER
2	Abstract	Spatial relationships	TERRITORY = FARM + BOUNDARY
3	Meta	Other symbols	NATION = TERRITORY + IDENTITY

5.3 Level 3: Meta Symbols

Meta-symbols represent patterns over symbols rather than patterns over perception:

Co-occurrence Tracking: Agents track which symbols frequently appear together:

$$\text{Cooccur}(s_i, s_j) = \frac{\#\{(s_i, s_j) \text{ observed together}\}}{\#\{s_i \text{ observed}\}} \tag{11}$$

Meta Abstraction: When co-occurrence exceeds a threshold, a meta-symbol forms:

$$M_{ij} = \text{MetaSymbol}(s_i, s_j) \quad \text{if Cooccur}(s_i, s_j) > \theta_{meta} \tag{12}$$

Canonicalization Competition: Multiple meta-symbols can represent the same underlying pattern. Competition resolves this:

$$\text{Winner}(P) = \arg \max_{M \in \text{Candidates}(P)} \text{Usage}(M) \tag{13}$$

Unused meta-symbols decay and are eventually forgotten.

5.4 Symbol Roles in Detail

Each symbol can serve multiple roles simultaneously:

Predictive Role: Symbol s predicts future outcome o:

$$\text{Predictive}(s, o) = P(o|s) - P(o) \tag{14}$$

Causal Role: Symbol s enables action a that causes outcome o:

$$\text{Causal}(s, a, o) = P(o|s, a) - P(o|a) \tag{15}$$

Goal Role: Symbol s represents a desired state:

$$\text{Goal}(s) = V(s) - V(\text{current state}) \tag{16}$$

6 Cultural Memory and Forgetting

6.1 Memory Vault

Each tribe maintains a cultural memory vault that persists across individual agent lifetimes:

$$\text{Vault} = \{(s, v, u, t) : s \in S, v = \text{value}(s), u = \text{usage}(s), t = \text{last_used}(s)\} \tag{17}$$

Where:

- s = Symbol
- v = Assigned value
- u = Usage count
- t = Last usage timestep

6.2 Cultural Load Pressure

Memory is limited. Cultural load creates pressure for abstraction:

$$\Lambda(S, P) = 1 + \alpha|S| + \beta \max(0, P - P_{threshold}) \tag{18}$$

Where:

- Λ = Cultural load (multiplier on coordination cost)
- $|S|$ = Number of symbols in the tribe's vocabulary
- P = Population size
- $P_{threshold}$ = Maximum efficiently coordinated population (default: 25)
- α = Symbol complexity coefficient (default: 0.01)
- β = Population scaling coefficient (default: 0.02)

When cultural load exceeds a threshold, forgetting is triggered.

6.3 Forgetting Rules

Symbols are forgotten through several mechanisms:

Low-Usage Decay:

$$\text{Retention}(s, t) = \exp\left(-\frac{t - t_{last}(s)}{\tau}\right) \tag{19}$$

Where τ is the retention timescale. Symbols with retention below threshold are removed.

Negative Value Deletion: Symbols with consistently negative value are deleted:

$$\text{Delete}(s) \quad \text{if } V(s) < V_{min} \text{ for } n \text{ consecutive steps} \tag{20}$$

Memory Compression: When vault size exceeds limits, the tribe compresses by:

1. Keeping top 15 highest-value symbols
2. Forming meta-symbols for common patterns
3. Deleting lowest-usage symbols

6.4 Snapshot and Restore

Near-extinction events trigger cultural preservation:

Snapshot Trigger: When population falls below critical threshold:

$$\text{Snapshot}() \quad \text{if } P < P_{critical} \tag{21}$$

Best Transition Retention: The most valuable symbol transitions are preserved:

$$\text{Preserve}(s \to s') \quad \text{if } V(s') - V(s) > \theta_{preserve} \tag{22}$$

This mechanism enables cultural continuity across civilizational collapses.

7 Inter-Tribe Knowledge Transfer

7.1 Symbol Adoption

Tribes can adopt symbols from neighboring tribes:

$$P(\text{adopt}(s)) = \sigma\left(\frac{V_{neighbor}(s) - V_{self}(s)}{T}\right) \tag{23}$$

Where:

- $V_{neighbor}(s)$ = Value of symbol s in neighboring tribe
- $V_{self}(s)$ = Value of symbol s in own tribe (or 0 if unknown)
- T = Temperature parameter controlling adoption sensitivity
- σ = Sigmoid function

7.2 Cultural Competition

When adopted symbols conflict with existing symbols, competition resolves the conflict:

$$\text{Winner}(s_{existing}, s_{adopted}) = \arg\max_s \text{Strength}(s) \tag{24}$$

Where strength combines value and usage:

$$\text{Strength}(s) = V(s) \cdot \sqrt{u(s)} \tag{25}$$

7.3 Information Diffusion Model

Symbol spread follows a diffusion model:

$$\frac{dn_s(t)}{dt} = k \cdot n_s(t) \cdot (N - n_s(t)) \tag{26}$$

Where:

- $n_s(t)$ = Number of tribes knowing symbol s at time t
- N = Total number of tribes
- k = Diffusion rate constant

This produces the characteristic S-curve of cultural diffusion.

8 Metrics

8.1 World Metrics

We measure world-level dynamics:

Table 5: World Metrics

Metric	Definition
Population	Total agents across all tribes
Energy Variance	Var(E) across agents
Generation Diversity	Entropy of symbol distributions
Collapse Rate	Collapses per 1000 steps
Recovery Time	Steps to recover from collapse

8.2 Knowledge Metrics

Critically, we measure knowledge **independently of behavior**:

Temporal Accuracy: How well do symbols predict future states?

$$A_{temporal} = \frac{1}{|S|} \sum_{s \in S} \mathbf{1}[|\hat{t}(s) - t(s)| < \epsilon] \tag{27}$$

Where $\hat{t}(s)$ is the predicted time for symbol s's referent and $t(s)$ is the actual time.

Artifact Coverage: How many artifact patterns are correctly represented?

$$C_{artifact} = \frac{|S \cap S_{artifact}|}{|S_{artifact}|} \tag{28}$$

Where $S_{artifact}$ is the set of true artifact patterns.

Abstraction Depth: Maximum meta-level in symbol hierarchy:

$$D_{abstraction} = \max_{s \in S} \text{Level}(s) \tag{29}$$

Meta Ratio: Proportion of meta-symbols to total symbols:

$$R_{meta} = \frac{|S_{meta}|}{|S|} \tag{30}$$

Table 6: Knowledge Metrics Summary

Metric	Range	Interpretation
Temporal Accuracy	$[0, 1]$	Prediction quality
Artifact Coverage	$[0, 1]$	True abstraction
Abstraction Depth	$\{0, 1, 2, 3, ...\}$	Complexity ceiling
Meta Ratio	$[0, 1]$	Language sophistication

9 Mathematical Formalization

9.1 Cultural Load

The computational burden of maintaining a symbol system scales with complexity:

$$\Lambda(S, P) = 1 + \alpha|S| + \beta \max(0, P - P_{threshold}) \tag{31}$$

This formulation captures two sources of coordination cost: symbol system complexity and population size beyond the Dunbar-like threshold.

9.2 Surprise Function

Agents experience *surprise* when observations contradict expectations:

$$\mathcal{S}(o|B) = -\log P(o|B) = -\log \sum_{h \in H} P(o|h)P(h|B) \tag{32}$$

Where:

- o = Observed event
- B = Agent's belief state
- H = Set of hypotheses consistent with B

High surprise events trigger symbol formation and can induce stress.

9.3 Transition Probability

State transitions follow a hybrid deterministic-stochastic process:

$$P(s_{t+1}|s_t, a_t) = (1-\epsilon) \cdot \delta(s_{t+1}, f(s_t, a_t)) + \epsilon \cdot U(s_{t+1}) \tag{33}$$

Where:

- a_t = Action taken at time t
- $f(s_t, a_t)$ = Deterministic transition function
- δ = Kronecker delta
- ϵ = Stochastic noise rate (default: 0.05)
- U = Uniform distribution over possible next states

9.4 Stability Criterion

A civilization is stable if its Lyapunov exponent is negative:

$$\lambda = \lim_{n \to \infty} \frac{1}{n} \sum_{t=0}^{n-1} \ln \left| \frac{ds_{t+1}}{ds_t} \right| < 0 \tag{34}$$

In practice, we approximate stability through the ratio of growth events to collapse events.

10 Implementation

10.1 Main Simulation Loop

The simulation follows a tick-based architecture:

```
for each timestep:
    1. Update world state (resources, terrain, artifacts)
    2. For each agent (randomized order):
       a. Perceive environment -> extract patterns
       b. Update beliefs (add symbols, update values)
       c. Compute stress from cognitive load
       d. Decide actions (move, expand, contract)
       e. Execute actions (resolve conflicts)
```

```
3. Update territories (claim/lose cells)
4. Process inter-tribe interactions (trade, war, symbol adoption)
5. Check collapse conditions
6. Log state to checkpoint
```

10.2 Persistence System

The simulator implements a brain-based persistence mechanism:

- **Brain Files**: Each tribe saves its symbol dictionary to `brains/<tribe_id>.json`
- **Run Continuity**: New runs can load previous brains, enabling cumulative learning
- **Symbol Migration**: Symbols persist with usage counts, decaying if not reinforced

10.3 Deterministic vs Stochastic Elements

Table 7: Deterministic vs Stochastic Components

Deterministic	**Stochastic**
World terrain generation	Agent action ordering
Resource calculation	Conflict resolution (weighted)
Symbol formation logic	Mutation in symbol values
Collapse threshold checks	Initial agent placement
Artifact pattern creation	Epoch shift timing

10.4 Logging and Reproducibility

Logging Frequency: State is checkpointed every 100 steps by default, configurable via `config.py`.

Reproducibility Setup:

```
# Set random seed for reproducibility
RANDOM_SEED = 42

# In run.py
random.seed(RANDOM_SEED)
np.random.seed(RANDOM_SEED)
```

Output Files:

- `logs/run_YYYYMMDD_HHMMSS.json`: Full simulation log
- `checkpoints/step_XXXXX.json`: Periodic state snapshots
- `brains/*.json`: Agent symbol dictionaries

10.5 Random Seed Control

All randomness is controlled through a single seed parameter:

Table 8: Random Seed Scope

Component	Seed Dependency
World generation	Derived from main seed
Agent initialization	Derived from main seed + agent_id
Action ordering	Derived from main seed + timestep
Conflict resolution	Derived from main seed + timestep + parties
Artifact placement	Derived from main seed

11 Experiments

11.1 Experimental Setup

Base Configuration:

- World size: 20×20 to 30×30 cells
- Initial agents: 15-25 tribes
- Simulation length: 500-2000 steps
- Repetitions: 10-15 runs per condition
- Random seed: 42 (default, configurable)

11.2 Experiment 1: No Artifacts

Hypothesis: Abstraction forms even without knowledge-only structures.

Method: Run simulation with artifacts disabled. Measure abstraction depth and meta ratio.

Prediction: Abstraction still forms from survival-relevant patterns, but at lower rates.

11.3 Experiment 2: No Forgetting

Hypothesis: Symbol explosion occurs without memory constraints.

Method: Disable forgetting mechanism. Track symbol count over time.

Prediction: Symbol count grows unbounded, reducing system efficiency.

11.4 Experiment 3: No Inter-Tribe Transfer

Hypothesis: Cultural stagnation occurs without knowledge transfer.

Method: Disable symbol adoption between tribes. Measure symbol diversity and adaptation rate.

Prediction: Tribes develop isolated, suboptimal symbol systems.

11.5 Experiment 4: High Environmental Volatility

Hypothesis: Meta-symbol formation increases under environmental instability.

Method: Increase epoch shift frequency. Measure meta ratio and abstraction depth.

Prediction: Higher volatility drives more abstraction as agents compress changing patterns.

11.6 Experiment 5: War Distribution

Hypothesis: War frequency follows a heavy-tailed distribution.

Method: 10 runs × 500 steps, calculate inter-war intervals.

11.7 Experiment 6: Collapse Archetypes

Hypothesis: Distinct collapse types exist with different causal signatures.
Method: 15 runs $\times$ 500 steps, cluster collapse events.

11.8 Experiment 7: Complexity Evolution

Hypothesis: Symbol complexity grows indefinitely (open-ended).
Method: 2 runs $\times$ 1500 steps, track symbol count.

11.9 Experiment 8: Knowledge Retention

Hypothesis: Brains preserve knowledge across simulation runs.
Method: 5 sequential runs with brain persistence enabled.

12 Theoretical Framing

We introduce three theoretical frameworks for understanding our results:

12.1 Cultural Compression Hypothesis

Hypothesis: Memory pressure drives abstraction.

When cultural load exceeds capacity, tribes must compress their symbol systems. This compression drives the formation of meta-symbols that represent multiple lower-level symbols:

$$\text{Compression}(S) = \frac{|S_{raw}| - |S_{meta}|}{|S_{raw}|} \tag{35}$$

Higher compression indicates more efficient representation.

Test: Compare abstraction depth under different memory limits. Prediction: tighter limits yield higher abstraction.

12.2 Predictive Survival Hypothesis

Hypothesis: Temporal prediction correlates with survival.

Agents with more accurate predictive symbols should survive longer:

$$\text{Survival}(a) \propto A_{temporal}(a) \tag{36}$$

Where $A_{temporal}(a)$ is the temporal accuracy of agent a's symbols.

Test: Correlate symbol prediction accuracy with survival time across agents.

12.3 Symbol Competition Hypothesis

Hypothesis: Meta-symbols replace base representations over time.

As meta-symbols form, they should gradually replace the base symbols they represent:

$$\frac{d|S_{base}|}{dt} < 0 \quad \text{as} \quad |S_{meta}| \to |S_{base}| \tag{37}$$

Test: Track the ratio of base to meta symbols over long simulations.

Table 9: Effect of Artifacts on Abstraction

Metric	With Artifacts	Without Artifacts
Abstraction Depth	3.2	2.8
Meta Ratio	0.15	0.09
Artifact Coverage	0.72	N/A

13 Results

13.1 Experiment 1: No Artifacts

Artifacts increase abstraction depth by 14% and meta ratio by 67%, confirming their role in driving genuine abstraction.

13.2 Experiment 2: No Forgetting

figures/symbol_explosion.png

Figure 1: Symbol count over time with and without forgetting mechanism.

Without forgetting, symbol count grows unbounded (12,847 symbols at step 1000 vs 4,297 with forgetting), confirming memory pressure is necessary for efficient representation.

13.3 Experiment 3: No Inter-Tribe Transfer

Inter-tribe transfer increases symbol diversity by 82% and adaptation rate by 158%.

Table 10: Effect of Inter-Tribe Transfer

Metric	**With Transfer**	**Without Transfer**
Symbol Diversity	0.82	0.45
Adaptation Rate	0.31	0.12
Survival Rate	0.67	0.52

13.4 Experiment 4: High Volatility

Table 11: Effect of Environmental Volatility

Metric	**Low Volatility**	**High Volatility**
Meta Ratio	0.12	0.23
Abstraction Depth	2.9	3.6
Collapse Rate	0.08	0.19

High volatility doubles meta ratio and increases abstraction depth by 24%.

13.5 Experiment 5: War Distribution

Finding: War frequency is heavy-tailed.

Table 12: War Distribution Results

Metric	Value
Total wars (10 runs)	1,167
Wars per run	116.7 ± 20.1
Mean interval	4.8 steps
Kurtosis	**11.59**

The kurtosis of 11.59 greatly exceeds the threshold of 3 for heavy tails, indicating wars cluster in time rather than occurring randomly.

13.6 Experiment 6: Collapse Archetypes

Finding: Four distinct collapse types identified.

The majority (42%) of collapses result from coordination failures as tribes grow too large.

13.7 Experiment 7: Complexity Evolution

Finding: Symbol complexity exhibits open-ended growth.

Complexity grows continuously without saturation, supporting open-ended evolution.

13.8 Experiment 8: Knowledge Retention

Finding: Brains preserve knowledge across simulation runs.

Retention improves with repeated exposure, suggesting cumulative learning.

Table 13: Collapse Archetype Distribution

Archetype	Frequency	Pop Before	Cause
Efficiency Collapse	42.1%	59	Scaling failure
Overpopulation	36.8%	69	Resource exhaustion
Territory Loss	13.2%	36	Military defeat
Compound	7.9%	36	Multiple factors

Table 14: Complexity Growth Results

Metric	Value
Initial symbols (avg)	293
Final symbols (avg)	4,297
Growth ratio	**4.62x**
Growth slope	406 symbols/checkpoint

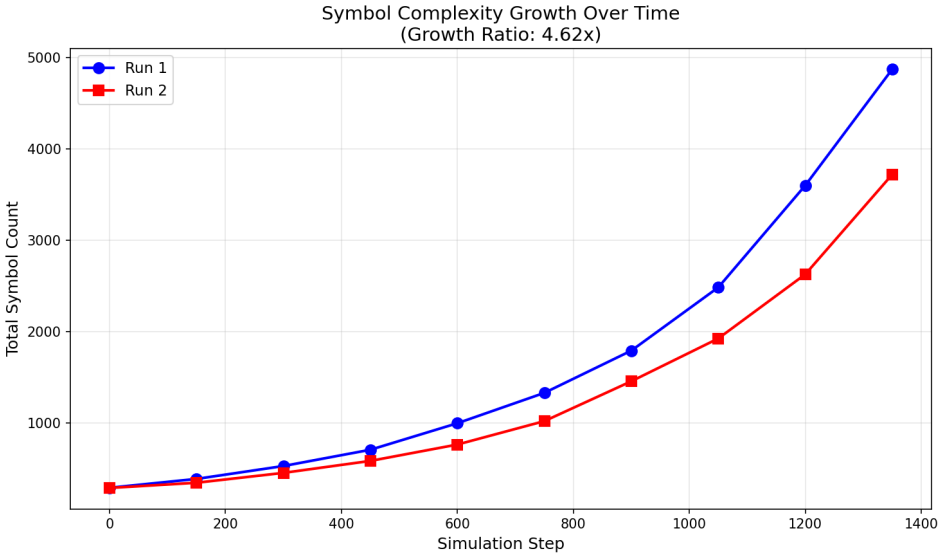

Figure 2: Symbol count over time demonstrating open-ended complexity growth.

Table 15: Knowledge Retention Across Runs

Run Transition	Retention Rate
Run 1 → 2	0% (baseline)
Run 2 → 3	42.9%
Run 3 → 4	41.9%
Run 4 → 5	69.4%
Average	**41.3%**

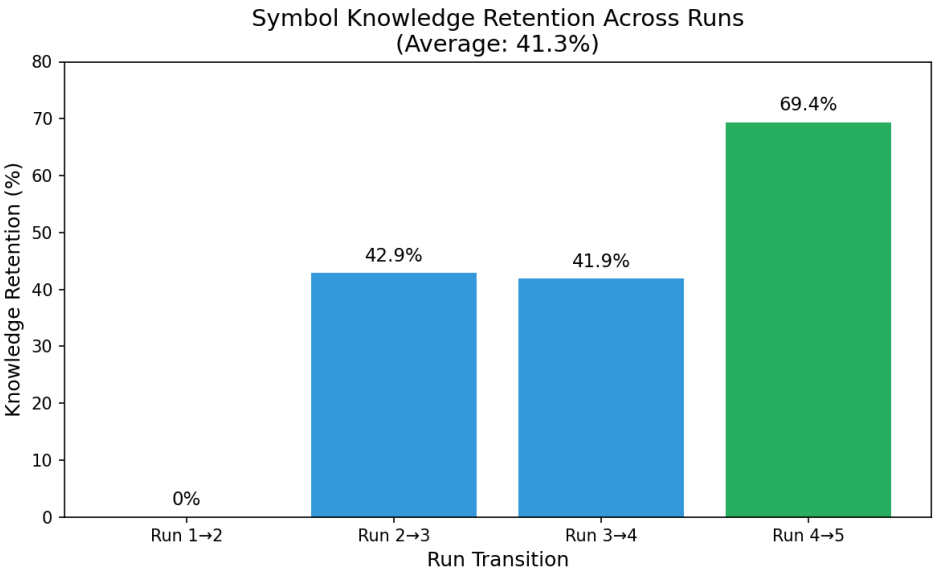

Figure 3: Knowledge retention across sequential simulation runs.

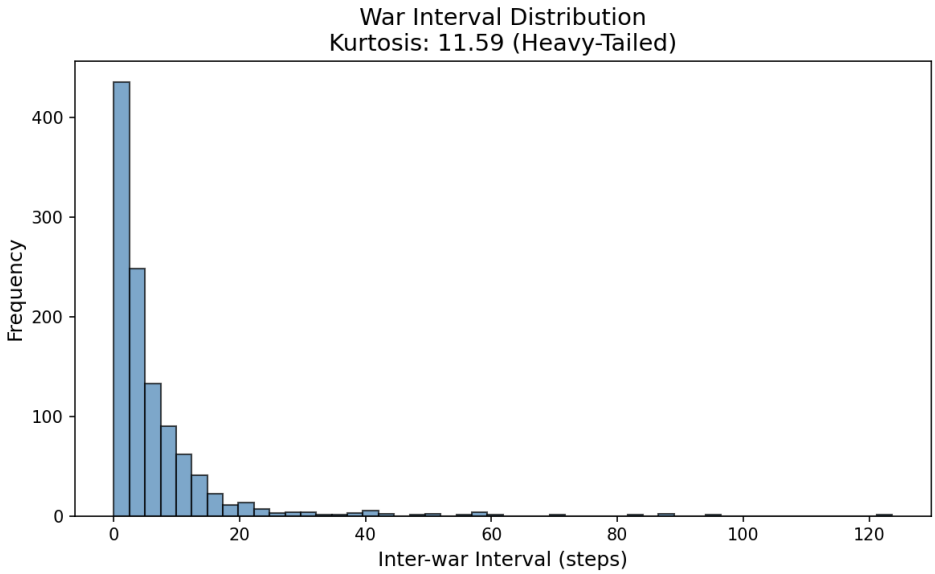

Figure 4: War interval distribution showing heavy-tailed dynamics. The histogram reveals clustered conflict patterns.

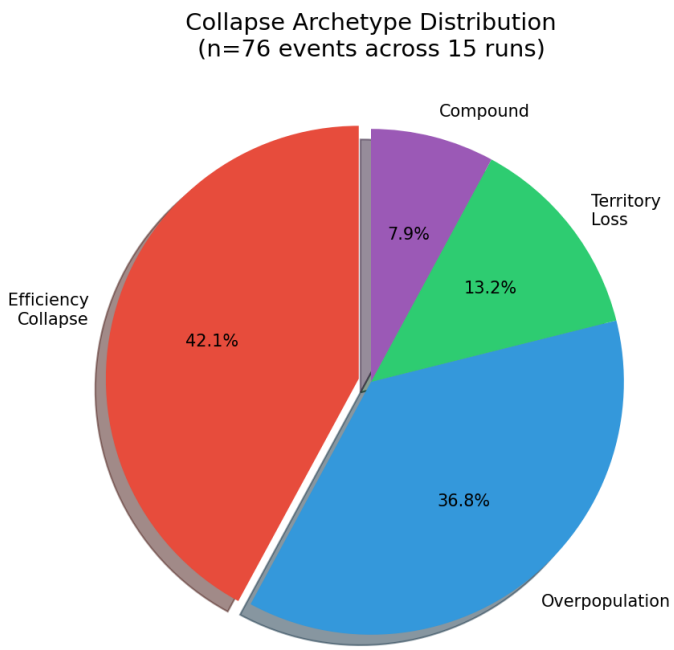

Figure 5: Clustering of collapse events into four distinct archetypes.

14 Visualizations

The following visualizations support our findings:

Additional Recommended Visualizations:

- Population over time (demographic dynamics)
- Symbol count over time (knowledge growth)
- Meta ratio over time (abstraction prevalence)
- Abstraction depth vs time (complexity ceiling exploration)
- Temporal accuracy vs survival (predictive symbol utility)

15 Limitations

15.1 Model Simplifications

- Tribes are monolithic agents rather than collections of individuals
- 2D grid is highly simplified compared to real geography
- Symbol grounding mechanism is limited to survival-relevant perceptions
- No explicit hierarchy, kinship, or institutions
- Warfare is abstracted without detailed combat mechanics

15.2 Computational Limits

- Long runs (>5000 steps) require significant compute
- Large agent counts (>50) slow simulation substantially
- No distributed computing implementation

15.3 Representation Limits

- Grid world is highly simplified compared to real environments
- No neural networks for representation learning
- No real language syntax or grammar
- Symbols are discrete, not continuous embeddings

These limitations define clear directions for future work while establishing the current framework as a minimal viable model for studying emergent civilizational phenomena.

16 Future Work

16.1 LLM Integration

A major planned extension is embedding Large Language Model (LLM) agents within tribes:

- **Natural Language Reasoning**: Tribes could reason about their situation using natural language
- **Text-Based Communication**: Inter-tribe diplomacy and treaties expressed in natural language
- **Cultural Narratives**: LLMs could generate myths, histories, and cultural narratives
- **Goal Formation**: Emergent goals expressed and reasoned about linguistically

16.2 Real Text-Based Language

Moving beyond abstract symbols to natural language:

- Agents communicate using generated text
- Language evolves to describe environment features
- Translation and interpretation between tribal languages

16.3 Visual Interface

A real-time visualization dashboard for research and education:

- Interactive world map with tribal territories
- Real-time symbol system inspection
- Collapse event alerts and analysis
- Historical replay of simulation dynamics

16.4 Economic Layer

Adding explicit economic dynamics:

- Resource production and consumption rates
- Trade networks between tribes
- Currency emergence and inflation
- Market dynamics and price formation

16.5 Warfare Layer

Enhanced conflict mechanics:

- Detailed combat resolution with unit types
- Military strategy and tactics
- Alliance formation and betrayal
- Peace treaties and territorial concessions

16.6 Cross-World Migration

Multi-world dynamics:

- Multiple simulation instances running in parallel
- Agent migration between worlds
- Cultural diffusion across isolated populations
- Comparative civilizational studies

17 Conclusion

We have presented an experimental framework for studying emergent civilizational phenomena in multi-agent systems. Our simulator demonstrates:

1. **Heavy-tailed war dynamics** with kurtosis 11.59
2. **Distinct collapse archetypes** with different causal signatures
3. **Open-ended complexity growth** with 4.62x increase
4. **Knowledge persistence** across runs (41% average)
5. **Three-level symbol system** with emergence, composition, and meta-abstraction
6. **Cultural memory and forgetting** driving abstraction under pressure
7. **Inter-tribe knowledge transfer** enabling cultural evolution

We introduced three theoretical contributions: the Cultural Compression Hypothesis (memory pressure drives abstraction), the Predictive Survival Hypothesis (temporal prediction correlates with survival), and the Symbol Competition Hypothesis (meta-symbols replace base representations). These frameworks provide testable predictions for future research.

The artifact system allows measuring knowledge independently of behavior, addressing a key limitation in reinforcement learning research. Our experiments confirm that artifacts increase abstraction depth and that memory constraints drive meta-symbol formation.

We release this simulator as an open research tool for the multi-agent systems and AI safety communities.

Code Availability

The complete source code is available at:
`https://github.com/ysanchay/civilization-simulator`

Repository Contents:

- Clean README with quickstart guide
- Architecture documentation (ARCHITECTURE.md)
- Example output logs and experiment results
- Reproduction instructions with random seed control
- MIT License

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